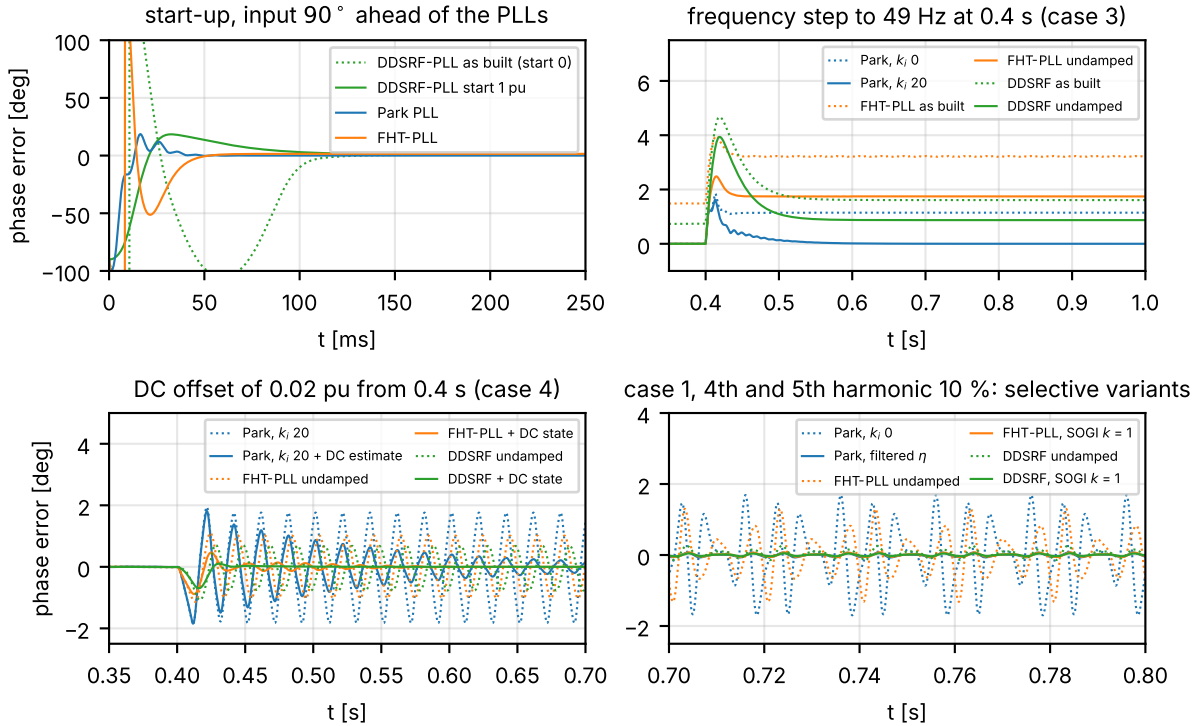

start-up, input 90 ° ahead of the PLLs
phase error [deg]
t [ms]
DDSRF-PLL as built (start 0)
DDSRF-PLL start 1 pu
Park PLL
FHT-PLL
frequency step to 49 Hz at 0.4 s (case 3)
t [s]
Park, k_i 0
Park, k_i 20
FHT-PLL as built
FHT-PLL undamped
DDSRF as built
DDSRF undamped
DC offset of 0.02 pu from 0.4 s (case 4)
phase error [deg]
t [s]
Park, k_i 20
Park, k_i 20 + DC estimate
FHT-PLL undamped
FHT-PLL + DC state
DDSRF undamped
DDSRF + DC state
case 1, 4th and 5th harmonic 10 %: selective variants
t [s]
Park, k_i 0
Park, filtered η
FHT-PLL undamped
FHT-PLL, SOGI k = 1
DDSRF undamped
DDSRF, SOGI k = 1